Assessment Test - 1

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CSA0519 - DataBase Management System

AT - 1

① Vechicle Management System - EER Diagram

(a) Identify the Supertype and Subtypes

Supertype: Vechicle

Subtype : Car and Truck

Vehicle is the supertype that contains common attributes. Car and Truck are Specialized entities that inherit the common attributes of Vehicle through an ISA relationship.

(b) Common attributes of Vehicle:-

1. Vehicle_ID → Primary Key
2. Registration_Number
3. Manufacturer
4. Model
5. Year
6. Purchase_Price

(c) Specialized Attributes of Car and Truck

Car Attributes:-

No_of_Door

No_of_fuel_type

Seating_Capacity

Truck Attributes

Load_Capacity

Truck_type

No_of_Axles.

Primary - key and Inheritance

Vehicle - ID is the primary key of vehicle supertype. Both Car and Truck inherit this primary key along with the common attributes of vehicle.

ISA Specialization

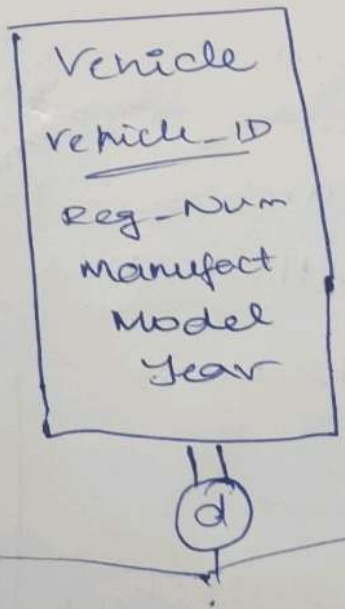
- Car inherit all common attributes from vehicle
- Truck inherit all attributes from vehicle.

Disjoint or Overlapping

* vehicle can belong to either Car or Truck subtype * Disjoint is

Total or partial

. The specilisation is total



Bank Management System:-

① The major entities are customer, Account, Branch, Employee, Transaction Loan, Loan - Payment

② & ③

Attributes and Primary Keys

Customer : Customer_ID, customer name, Address, Phone, Email.

Account : Account_NO

Branch : Branch_ID

Employee : Employee_ID

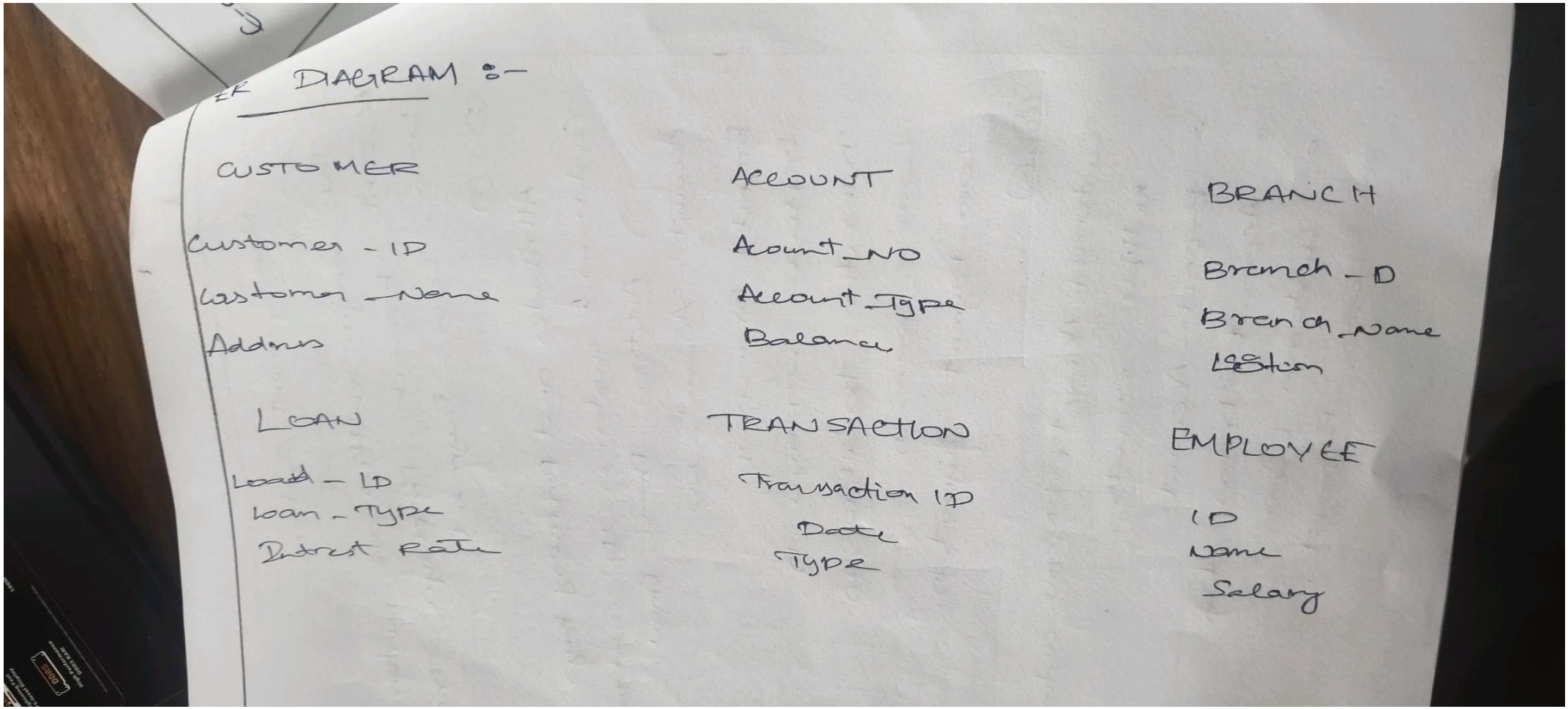
Transaction : transaction_ID

Loan : Loan_ID

Loan_Payment : Payment_ID.

④ & ⑤

- Customer - Account : 1:N ; one customer may hold multiple accounts
- Branch - Account.
- Account Transaction
- Customer Loan
- Loan Payment.


ER DIAGRAM :-
CUSTOMER
Customer - ID
Customer_Name
Address
ACCOUNT
Acount_NO
Account_Type
Balance
BRANCH
Branch_ID
Branch_Name
Location
LOAN
Loan_ID
Loan_Type
Interest Rate
TRANSACTION
Transaction ID
Date
Type
EMPLOYEE
ID
Name
Salary

E-commerce Website:

① Major Entities

* Customer, Products, Category, order, payment and Delivery

② Attributes and Primary key

- Customer : Customer_ID
- Product : Product_ID
- Category : Category_ID
- Delivery : Delivery_ID

③ Order_Product Relationship

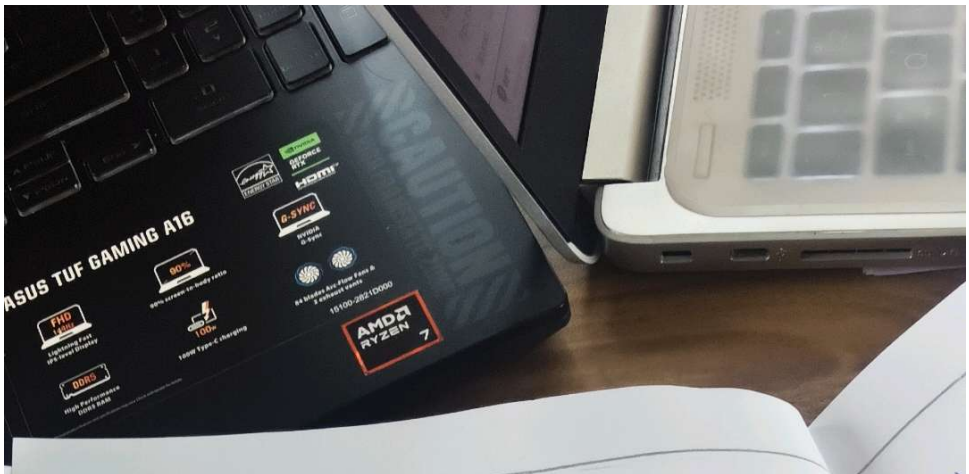
* Order and Product have many to many relationship

The attributes Quanty and Unit Price are stored in Order_Item

④ Product is supertype

Subtypes are

- Electronics
- Clothing
- Grocery



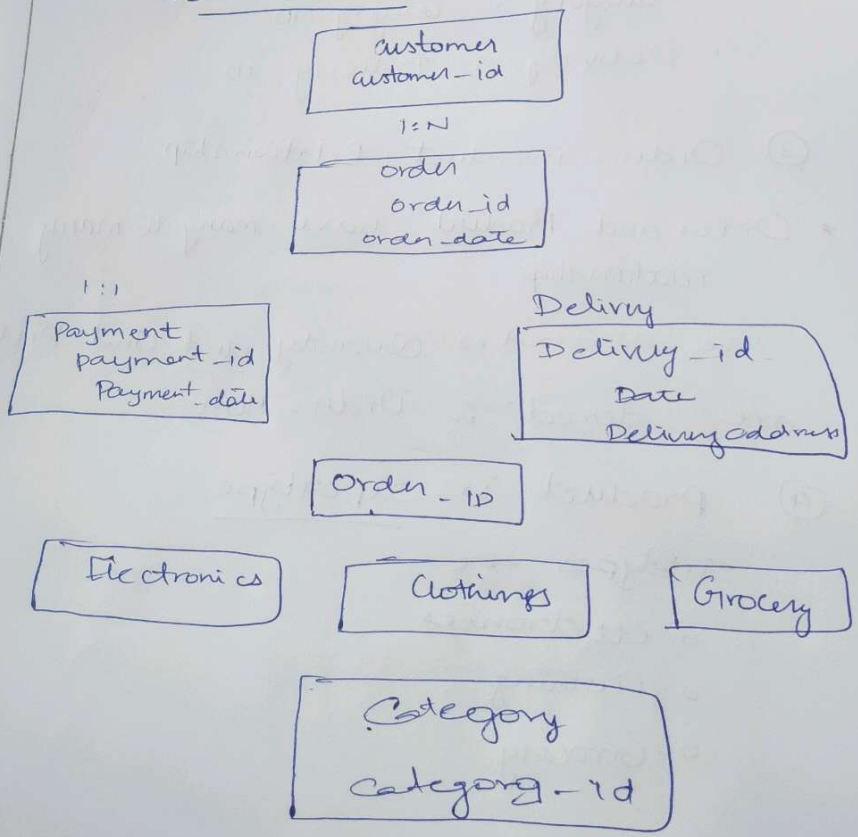
6. Specialization Constrains

- Disjoint → only belongs three subtypes
- TOTAL → every product belongs to three subtype.

7. & Relationship and Cardinalities

Customer - order → 1:N
order - payment → 1:1
order - Delivery → 1:1

9- EER DIAGRAM



subtypes

① Hospital Management System

1. Major entities

. The major entities are Staff, Doctor, Nurse, Admin, Department, Appointment.

2-4 Staff Specialisation and attributes

Staff	Doctor
Staff_id	◦ Staff_id
Name	◦ Specilisation
Address	◦ License_Number
Phone	
Salary	

5-7 Week entity and Identifying Relationship

8-9 Cardinality

Department

Dept - Doctor	1:N
Patient - Appoin	1:N
Doctor - Appoin	1:N
Patient_Medical_Record	1:N

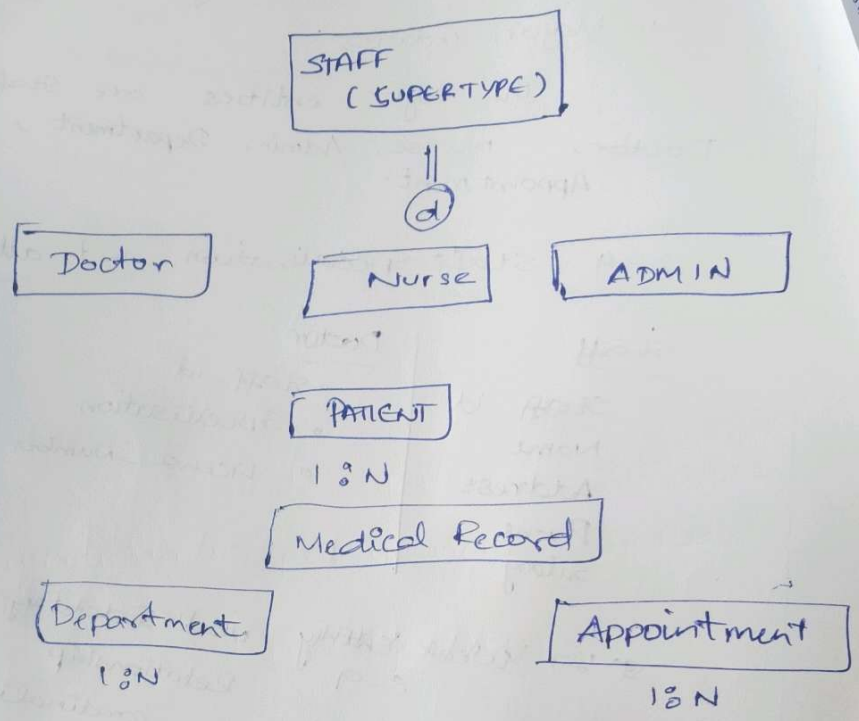
5-7 Week entity and Identifying Relation

Medical record is week entity

partial key - Record_Rate

Identifying relationship - HAS_RECOR

EER DIAGRAM



PATIENT

1 : N

Medical Record

Department

1 : N

Appointment

1 : N

Database Administrator

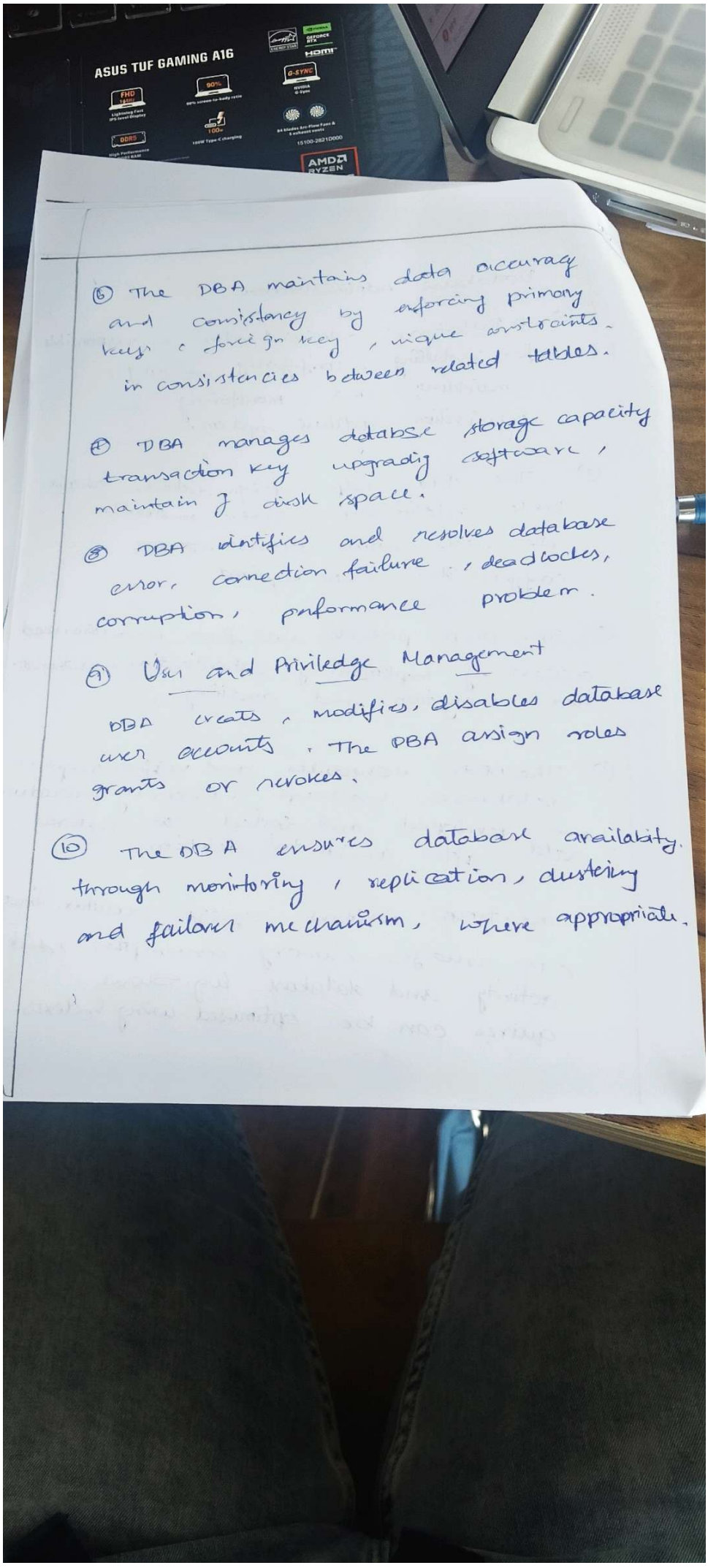
① A Database administrator is responsible for installing, configuring, securing, maintaing and monitoring an organisation database system.

② The DBA helps design database schemas table, relationship, indexes, constraints. DBA selects suitable database software configures it. when required.

③ The DBA protects data from unauthorized access by implementing athentication, authorization, encryption and auditing.

④ The DBA schedules and verifies regular database backup. Recovry procedure are developed and tested to restore data after accidental deletion.

⑤ The DBA monitors query execution time CPU usage, memory concemption, disk activity and database log. Slow quires can be optimised using indexes.



(6) The DBA maintains data accuracy and consistency by enforcing primary keys, foreign key, unique constraints. in consistencies between related tables.

(7) DBA manages database storage capacity transaction key upgrading software, maintain of disk space.

(8) DBA identifies and resolves database error, connection failure, deadlocks, corruption, performance problem.

(9) User and Priviledge Management

DBA creats, modifies, disables database user accounts. The DBA assign roles grants or revokes.

(10) The DBA ensures database availability through monitoring, replication, clustering and failover mechanism, where appropriate.